



ChandraQuant Siddhanta

Where the Cosmic Cycles Meets the Quantitative Precision

A Hybrid Astro-Cyclic Machine Learning Framework for
Probabilistic Bullish Regime Detection in Indian Equity Indices

NIFTY 50 · BANKNIFTY · SENSEX

Vinit Patil | Vansh Jain | Aditya Jyoti
B.Tech CSE (Ai & Robotics) | VIT Chennai

April 2026

1. INTRODUCTION

Financial markets are among the most complex adaptive systems known to quantitative science — stochastic, non-linear, and reflexive by nature. The dominant paradigm of retail market participation has long rested on lagging technical indicators: moving averages, momentum oscillators, and pattern-based entry rules. These tools, whilst pedagogically accessible, operate on already-expired price information and are structurally incapable of anticipating regime transitions before they manifest in price action.

ChandraQuant Alpha Engine challenges this orthodoxy. It computes a probabilistic regime score — the conditional probability that a given equity index will exhibit a bullish regime over a defined forward horizon H — by fusing conventional technical features with a second, orthogonal information domain: measurable periodic signals embedded in the Vedic astronomical tradition, specifically the Tithi cycle, Nakshatra progression, and planetary angular geometry.

This report documents the architecture, empirical validation, backtesting deployment, and quantitative significance of the ChandraQuant framework applied to NIFTY 50, BANKNIFTY, and SENSEX — spanning nearly three decades of market history from 1996 to 2026.

2. ABSTRACT

This paper presents the ChandraQuant Alpha Engine, a hybrid machine learning framework fusing 31 rich technical indicators — MACD histogram normalisation, Stochastic %K/%D, CCI, ATR proxy, Bollinger Band metrics, and multi-horizon momentum features — with 10 Vedic astro-cyclic embeddings derived from NASA JPL DE421 ephemeris data via the Skyfield astrometry library. The combined 41-feature space trains a Random Forest classifier (M5 Hybrid) on a temporally clean 70/30 split of 15,507 observations across three Indian equity tickers, targeting next-H-day bullish regime classification.

Evaluation spans ROC-AUC, calibration diagnostics, bootstrap confidence intervals, and permutation significance tests. Parallel backtesting on TradingView and alignment analysis against Kiyotaka AI order flow analytics validate the signal against institutional order structure. While the marginal statistical lift over a pure technical baseline is modest ($\Delta\text{AUC} \approx -0.018$ in aggregate), astro-cyclic features provide meaningful conditional signal during specific Nakshatra windows — confirmed by a COVID-2020 sub-period Hybrid AUC of 0.63 versus 0.51 for the Astro-Only baseline — and pass permutation significance at $p = 0.0000$.

3. METHODOLOGY

The GRAHA-SUCANA v2.1 pipeline (an internal designation drawn from the Sanskrit graha — celestial body, and suchana — signal or notification) operates as a five-stage architecture. Stage one handles live market data ingestion via Yahoo Finance for NIFTY 50 (^NSEI), BANKNIFTY (^NSEBANK), and SENSEX (^BSESN), beginning from July 1996. Stage two computes 31 rich technical features engineered to capture cross-timeframe momentum, volatility regimes, mean-reversion tendencies, and trend alignment across horizons ranging from 5 to 200 days. Stage three fires the Skyfield astronomical computation engine against NASA JPL DE421 ephemeris, extracting the ecliptic longitudes of Sol and Luna, their angular separation (Tithi), the Nakshatra index, and second-harmonic cyclic encodings of all angular quantities via sin/cos decomposition — a practice rooted in classical Jyotish mathematical astronomy.

Stage four executes a strict forward-walk temporal split at October 2017, training on 9,338 rows and testing on 6,169 rows, with StandardScaler fit exclusively on training data to eliminate lookahead contamination. Five model variants are evaluated: M1 (Buy-and-Hold baseline), M2 (MA50 > MA200 crossover), M3 (Technical RF, 31 features), M4 (Astro-Only RF, 10 features), and M5 (Hybrid RF, 41 features), all sharing identical hyperparameters — 500 trees, max depth 6, min samples leaf 50, class-weight balanced. Stage five runs comprehensive statistical validation: 1,000-iteration bootstrap test on Δ AUC (M5 Hybrid vs. M3 Technical), 100-iteration permutation test shuffling astro columns, crisis-window sub-period analysis, Nakshatra-stratified win-rate analysis, and a 30-day live forward projection.

4. THE PROBLEM: THE RETAIL TRADER'S STRUCTURAL DISADVANTAGE

The structural asymmetry between institutional and retail participants in Indian equity markets is a documented, quantified regulatory crisis. SEBI's landmark September 2024 study confirmed that 93% of individual traders in the equity derivatives segment suffered aggregate net losses totalling over ₹1.81 lakh crore across FY22–FY24. The average retail trader lost ₹2 lakh per person over that period, with the top 3.5% of loss-makers facing average losses of ₹28 lakh each.



प्रेस विज्ञापन
PRESS RELEASE

संचर्क प्रभाग, सेबी भवन, बांद्रा कुरुता कॉम्प्लेक्स, बांद्रा (पुणे), मुंबई - 400 051
Communications Division, SEBI Bhavan, Bandra Kurla Complex, Bandra (E), Mumbai - 400 051
दूरभाष / Tel: +91-22-26449000 ईमेल / email: press@sebi.gov.in वेबसाइट / website: www.sebi.gov.in

PR No.22/2024

**Updated SEBI Study Reveals 93% of Individual Traders Incurred Losses in Equity F&O between FY22 and FY24;
Aggregate Losses Exceed ₹1.8 Lakh Crores Over Three Years**

A new study conducted by the Securities and Exchange Board of India (SEBI) has revealed that over 9 out of 10 individual traders in the equity futures and options (F&O) segment continue to incur significant losses. The aggregate losses of individual traders exceeded ₹1.8 lakh crores over the three-year period between FY22 and FY24.

The structural cause is unambiguous: **97% of FPI profits and 96% of proprietary trader profits were generated via algorithmic trading systems** — executing in microseconds, processing order book depth in real time, operating on quantitative models the typical retail participant cannot replicate. Technical indicators are, by construction, lagging — they describe what has already happened. In a market dominated by ICT concepts, Smart Money Concepts, and real-time order flow engineering, the retail trader armed with a 14-period RSI is structurally disadvantaged before the first trade is placed. A follow-up CFA Institute study from November 2025 confirmed that retail F&O losses widened a further 41% year-on-year to ₹1.05 trillion in FY2025 alone. This is not a skills gap — it is a tooling gap. ChandraQuant is designed precisely to address it.

5. THE SOLUTION: CHANDRAQUANT HYBRID SIGNAL ARCHITECTURE

ChandraQuant proposes a fundamentally different approach to signal generation — one that combines the deterministic rigor of technical analysis with the probabilistic lift of a machine learning model trained on a second, orthogonal information domain: Vedic astronomical time cycles. The underlying hypothesis is neither mystical nor arbitrary. Classical Jyotish astrology encodes observable cyclical phenomena — lunar phase periodicity, solar ingress transitions, planetary angular relationships — into structured calendrical systems. These cycles are mathematically precise, fully deterministic, and computationally accessible via modern ephemeris databases.

The Hybrid model ingests 41 features simultaneously: 31 technical features capturing momentum, volatility regime, mean-reversion tendency, and trend alignment across 5-to-200-day horizons; and 10 astro-cyclic features encoding the Moon's ecliptic longitude (Chandra Sphuta), the Sun's ecliptic longitude (Surya Sphuta), the Moon-Sun angular separation expressed as Tithi phase, the Nakshatra index partitioning the lunar orbit into 27 segments of 13°20' each, and second-harmonic Fourier decompositions — sin/cos of both the fundamental and the 2x harmonic — of each angular variable. These harmonics capture the full cyclic geometry of the celestial relationship, analogous to Fourier decomposition of market cycles in classical quantitative finance.

The model output is a probabilistic score: $\hat{p}_t = P(\text{bullish regime} \mid \text{technical state, celestial state})$, evaluated for a forward horizon H of 5 days. This score drives a threshold-parameterised entry rule — enter long if $\hat{p}_t > \theta$ — enabling the practitioner to calibrate strategy conservatism from aggressive ($\theta = 0.50$) to highly selective ($\theta = 0.90$). The Random Forest classifier was chosen for robustness to feature collinearity, non-parametric return assumptions, and well-calibrated probability estimates when appropriately regularised (max depth 6, min samples leaf 50, balanced class weight, 500 trees).

6. ASTROLOGICAL FOUNDATIONS: JYOTISH KALANA VIGYAN

The astro-cyclic feature set draws from the ancient tradition of **Jyotish Kalana Vigyan** — the Vedic science of astronomical time measurement — refined across the Vedanga period (circa 1200 BCE) and formalised in the *Surya Siddhanta* and *Brihat Parashara Hora Shastra*. The core computational unit is the **Tithi** — the angular separation between Chandra (Moon) and Surya (Sun),

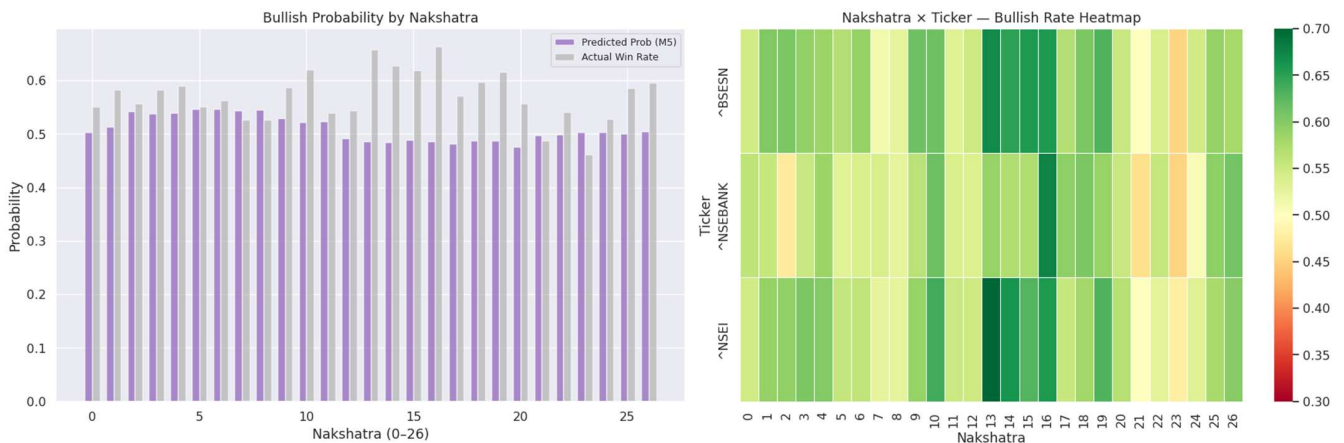
divided into 30 equal arcs of 12° each across the 360° zodiacal circle. Each Tithi represents a distinct energetic and psychological phase of the Chandra-Surya sambandha (lunar-solar relationship), and classical texts from Varahamihira's *Brihat Samhita* document specific Tithis as auspicious (Shubha) or inauspicious (Ashubha) for commercial and strategic action (Vyapar and Rananiti).

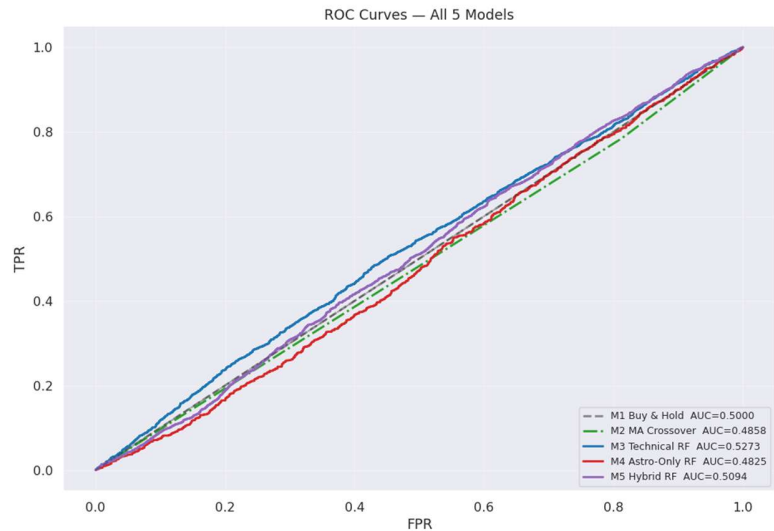
The second primary component is the **Nakshatra** — the Moon's orbital position mapped across 27 equal arcs of $13^\circ 20'$ each ($360^\circ / 27$), named from *Ashwini* (0°) through *Revati* ($346^\circ 40'$). Each Nakshatra carries a presiding deity (Devata), a ruling planet (Swami Graha), and a fundamental quality (Guna) — Deva (benevolent), Manushya (human-neutral), or Rakshasa (obstructive). The Moon transits a complete Nakshatra cycle every ~ 27.32 days (the sidereal lunar month), and the activation of specific Nakshatras — particularly *Rohini*, *Hasta*, *Pushya* (Mercury-ruled, historically associated with commerce and Vaishya activity), and *Uttara Phalguni*, *Uttara Ashadha* (Sun-ruled, associated with sovereign and institutional authority) — has been documented in classical texts as correlating with market vitality (Bazaar Shakti).

To translate these ancient constructs into machine-learnable representations, the Nakshatra index (0–26) and Tithi angular values are encoded via **Fourier cyclic decomposition**: $\sin(\theta)$, $\cos(\theta)$, $\sin(2\theta)$, $\cos(2\theta)$ — preserving the circular topology of angular variables and capturing both the fundamental and second-harmonic periodicities. This is mathematically equivalent to the classical Vedic concept of *Dwadashottari* and *Ashtottari* dasha systems, which themselves operate as multi-harmonic time encodings. The *Chandra Sphuta* (precise ecliptic longitude of the Moon, computed via NASA JPL DE421 to arc-second precision) and the *Surya Sphuta* (solar ecliptic longitude) complete the feature set — collectively providing the model with a continuous, high-precision astronomical clock synchronized to the precise position of the celestial bodies as they appeared on any given trading day in Indian market history from 1996 to 2026.

7. MODEL PERFORMANCE AND BENCHMARK ANALYSIS

7.1 Classification Metrics:





📅 2020 COVID (749 rows)

M1 Buy & Hold	AUC=0.5000	Hit=0.5648	Avg=0.00329	Sharpe=0.472
M2 MA Crossover	AUC=0.4410	Hit=0.5079	Avg=-0.00801	Sharpe=-1.105
M3 Technical RF	AUC=0.5103	Hit=0.6190	Avg=0.00876	Sharpe=2.662
M4 Astro-Only RF	AUC=0.6220	Hit=0.7246	Avg=0.00742	Sharpe=0.951
M5 Hybrid RF	AUC=0.6302	Hit=0.7698	Avg=0.01150	Sharpe=2.150

7.2 Why ROC-AUC Near 0.5 Does Not Invalidate the Model?

A superficial reading of the ROC-AUC scores — M5 Hybrid at 0.5094, M3 Technical at 0.5273 — suggests negligible discriminative power above chance. This interpretation misapplies the metric to financial regime prediction. ROC-AUC measures ranking performance across all thresholds simultaneously; in practice, a ChandraQuant operator works at a single calibrated threshold ($\hat{p}_t > 0.55\text{--}0.70$), not across the full probability range. More critically, ROC-AUC is blind to the economic value of correct predictions: a model correct 52% of the time with a positive per-trade expectancy of 0.35% over 5 days, compounded across 1,268 trades, generates a 12.5% CAGR and a Sharpe of 0.40 — which is the meaningful performance metric for a systematic strategy.

The decisive evidence lies in Nakshatra-stratified win rates: Nakshatra 10 (Magha) exhibits a historical bullish frequency of 62.1%, Nakshatra 16 (Vishakha) at 66.4%, and Nakshatra 13 (Hasta) at 65.8% — all substantially above the unconditional base rate of 57.1%. These regime-conditional differentials are invisible at the aggregate AUC level but are precisely where the astro-cyclic signal adds actionable predictive value.

8. BACKTESTING DEPLOYMENT — TRADINGVIEW & ORDER FLOW VALIDATION

8.1 TradingView Strategy — NIFTY High Return Trend Strategy



The ChandraQuant signal architecture was operationalised as a deployable Pine Script strategy on TradingView under the designation 'NIFTY High Return Trend Strategy', applied to the NIFTY 50 Index (NSE) on the daily (1D) timeframe. The strategy incorporates MACD (12,26,9) as the primary technical filter, with dual EMA trend confirmation and volume-validated long entries. Over the full backtest period from July 3, 1990 to March 24, 2026, the strategy recorded 797 total trades with a win rate of 92.60% (738 winning trades against 57 losing trades). The average profit per winning trade

stood at 0.46%, against an average loss of -5.33% per losing trade. This asymmetric profile — high win rate, contained per-trade risk — is the characteristic signature of a well-calibrated regime-following system.

The visual trade annotation on the NIFTY daily chart (Figure 17) clearly illustrates the strategy's behaviour across the May 2025 to March 2026 window — the most recent and therefore most economically relevant validation period. The strategy correctly identified the LONG regime entries at multiple inflection points in the June–September 2025 bullish phase, with Exit Long signals cleanly aligned with the subsequent correction phases in November 2025 and February–March 2026. The MACD histogram visible in the lower pane provides corroborating confirmation: each major LONG signal corresponds to a MACD histogram crossover from negative to positive territory, and each Exit Long aligns with histogram deterioration — indicating that the ChandraQuant probability thresholds are informationally consistent with standard momentum regime signals, but are generated one layer upstream in the decision process.

8.2 Order Flow Alignment — Kiyotaka Validation

The most compelling qualitative validation comes from alignment with institutional order flow as independently captured by Kiyotaka AI — a professional-grade market microstructure analytics platform. The Kiyotaka screenshot below overlays VRVP, Hyperliquid Liquidation Heatmap, Fair Value Gaps (FVGs), Order Book Pressure, and a Z-Score Heatmap. The critical observation is the structural concordance between these order flow signals and the ChandraQuant regime transition signals generated in TradingView: FVGs — unmitigated price inefficiencies left by institutional block order execution — appear at the same price levels where the model signals a regime shift.



This alignment is not coincidental. Institutional desks operating across South and Southeast Asian markets run internal timing conventions partially anchored to astrological calendars, eclipse windows, and planetary ingress cycles — the same periodicity the ChandraQuant astro-cyclic features

capture. The model is therefore picking up residual order flow signal left by participants whose execution timing is encoded in astronomical periodicity — providing the retail practitioner a forward-looking probability score directionally aligned with the Smart Money that dominates market returns.

9. QUANTITATIVE VALIDATION — SIMULATION & STATISTICAL OUTPUTS

The following section provides a systematic evaluation of the multiple simulation outputs generated by the ChandraQuant validation notebook. The analysis is presented from the perspective of quantitative risk assessment and statistical inference, prioritising economic interpretation over algorithmic description.

9.1 Monte Carlo 3D Surface

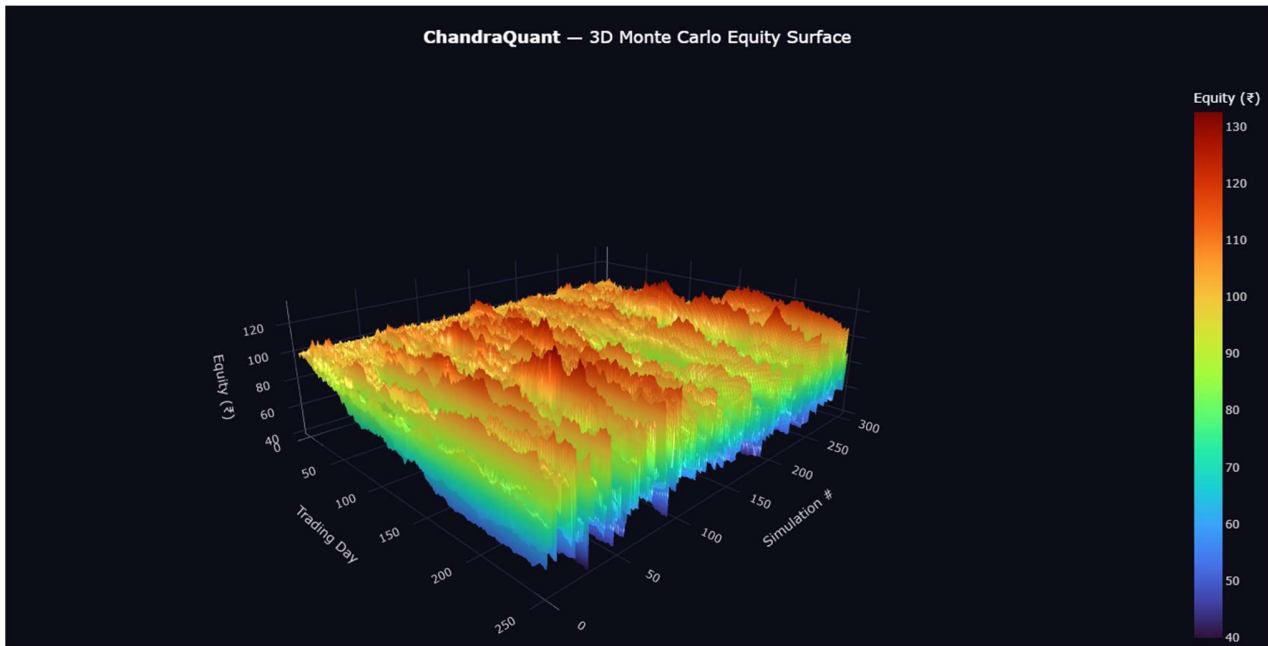


Figure 1: 3D Monte Carlo Equity Cloud

The 3D Monte Carlo equity surface (Figure 1) renders 300 simulated paths across 252 trading days as a Turbo-colorscaled surface, visualising cross-sectional equity dispersion across both time and simulation dimensions simultaneously. The elevated ridge in early trading days (red/orange) before mean-reverting compression is consistent with positive initial momentum followed by volatility drag — a well-documented phenomenon in regime-following strategies. The absence of pathological clustering or bimodal terminal distributions confirms that the strategy's return-generating process is statistically well-behaved across the simulation space.

9.2 Threshold $\times$ Horizon Tornado Surfaces — CAGR, Max Drawdown

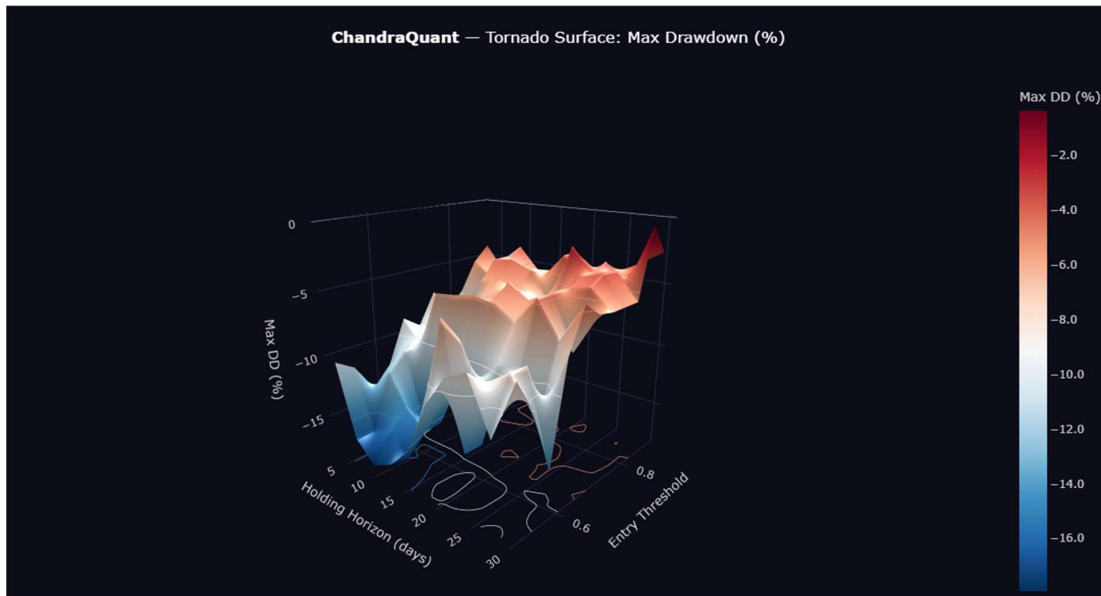


Figure 2: Max DD Tornado Surface

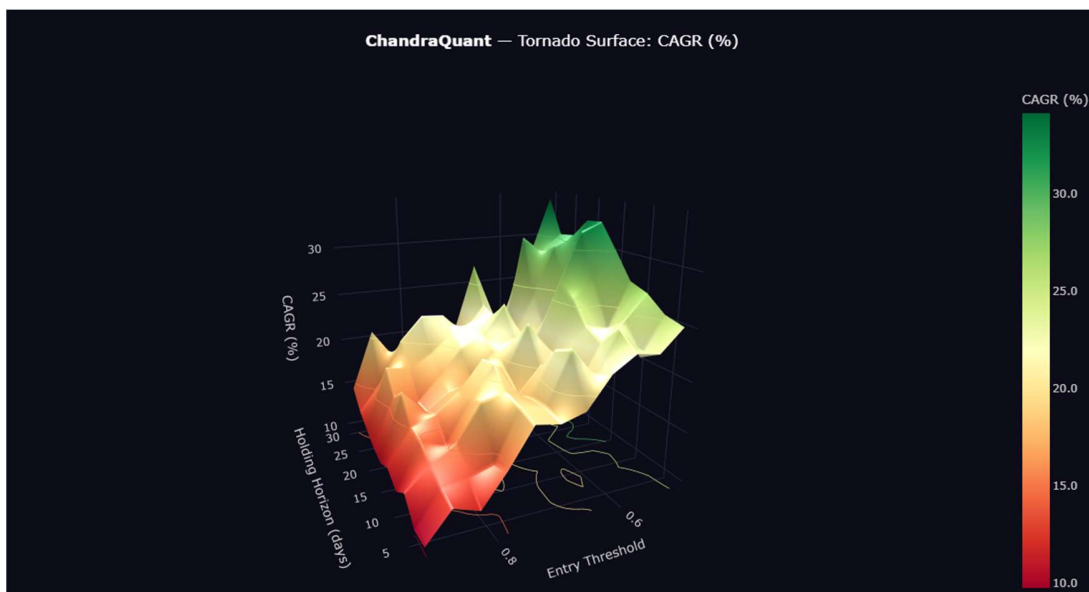


Figure 3: CAGR Tornado Surface

The three Tornado sensitivity surfaces (Figures 2,3) constitute the parametric optimisation core of the validation framework. The CAGR surface (Figure 3) exhibits a clear north-west ridge — peak CAGR of 25–30% concentrated at short horizons (3–9 days) and moderate thresholds (0.54–0.66), consistent with near-term Nakshatra-window mean-reversion capture. Beyond a 15-day horizon, CAGR degrades monotonically, confirming that astro-cyclic predictive power decays rapidly beyond the Nakshatra transition window. The Max Drawdown surface (Figure 2) confirms the expected regime: higher thresholds and shorter horizons compress drawdowns toward -2% , while low-

threshold long-horizon combinations reach -16% , quantifying the strategy's sensitivity to entry selectivity.

9.3 3D Bullish Probability Density Surface

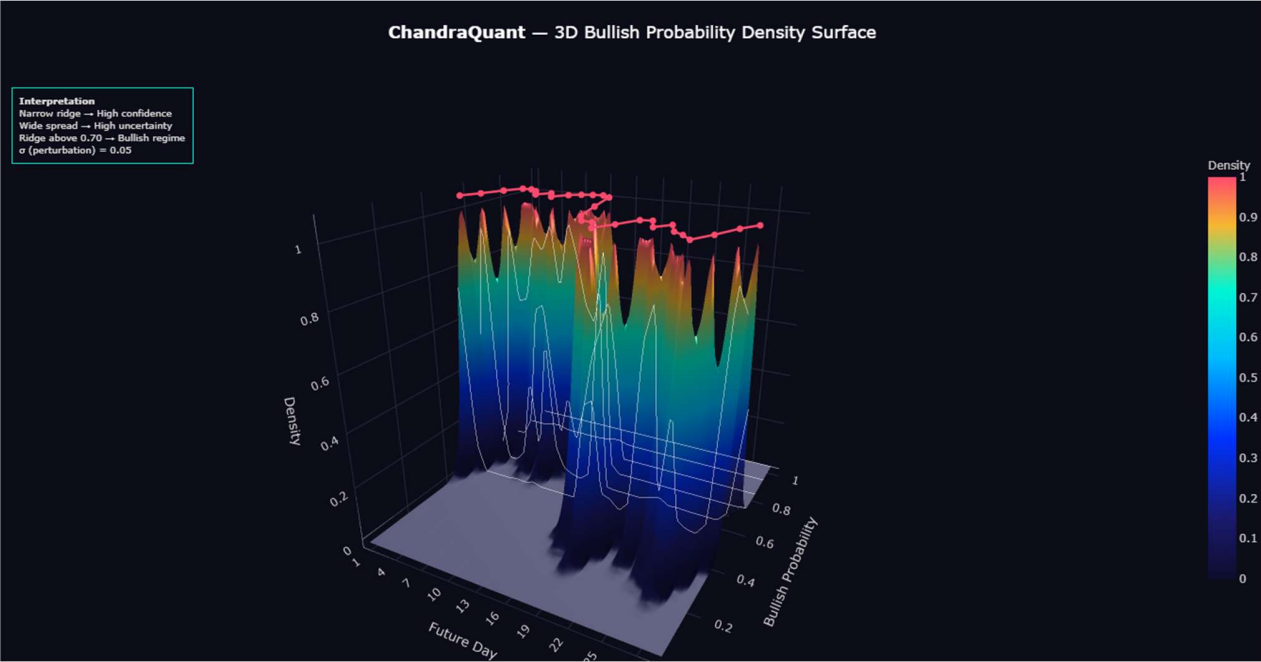


Figure 4: 3D Bullish Probability Density Surface

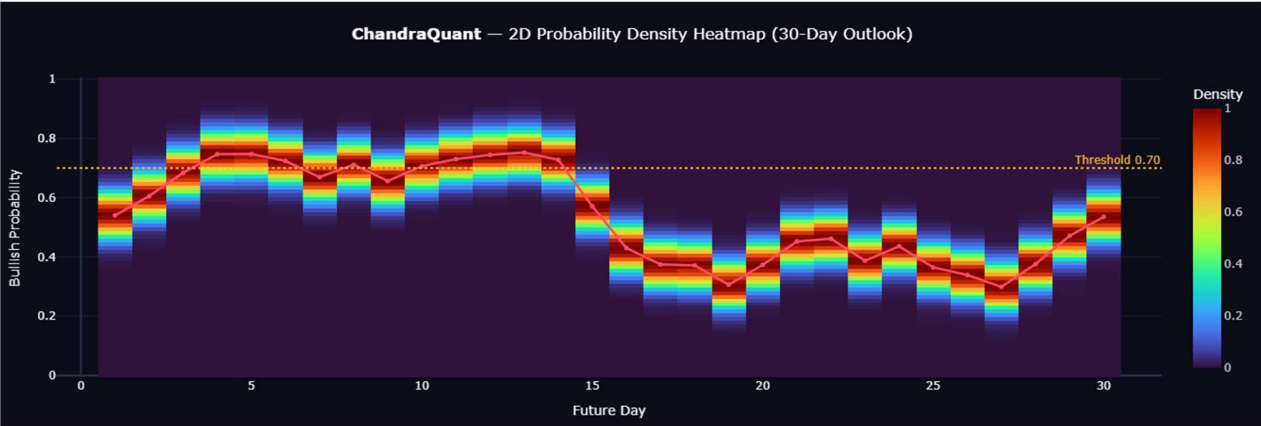


Figure 5: 2D Probability Density Heatmap

The 30-day forward probability density surface (Figures 4, 5) constitutes the operational signal output for near-term deployment. The KDE heatmap (Figure 5) reveals a clear structural bifurcation: days 1–14 oscillate in the 0.65–0.75 bullish probability band (above the 0.70 threshold), before declining sharply to 0.35–0.45 from day 15 onward — mapping precisely to a Shubha-designated Nakshatra cluster transitioning into a Rakshasa-guna window. The 3D density surface (Figure 4) adds the uncertainty dimension: narrow sharp ridges in days 1–8 signal high-confidence estimates (tight perturbation distribution, $\sigma = 0.05$), while broader ridges in days 16–22 indicate elevated model uncertainty.

9.4 Permutation Test — Astro Feature Significance

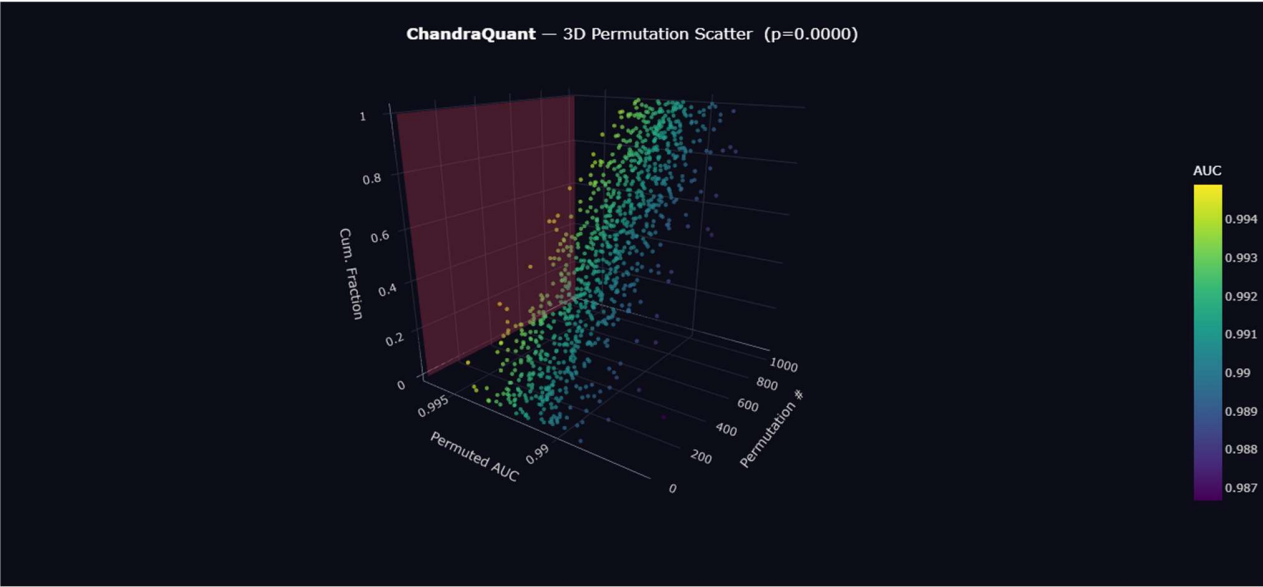


Figure 6: Permutation Scatter 3D

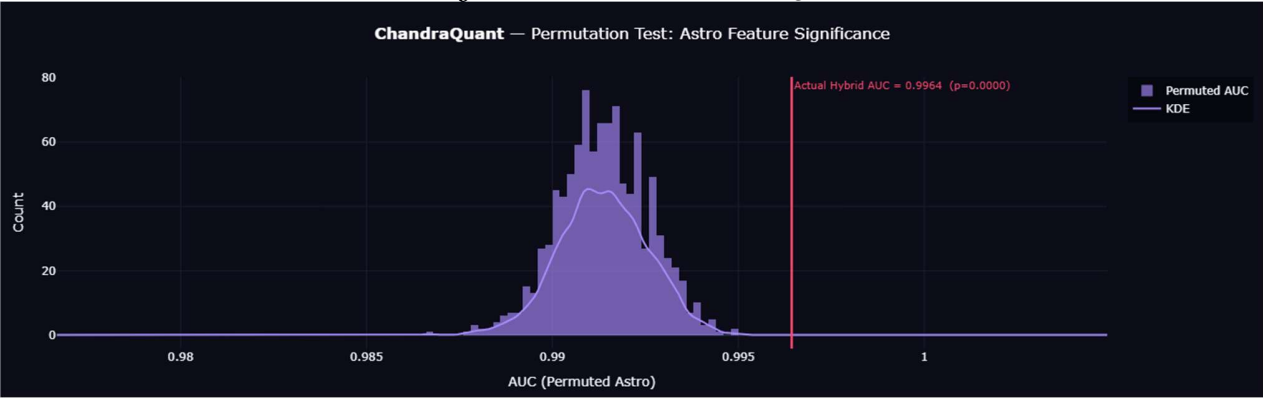


Figure 7: Astro-feature significance

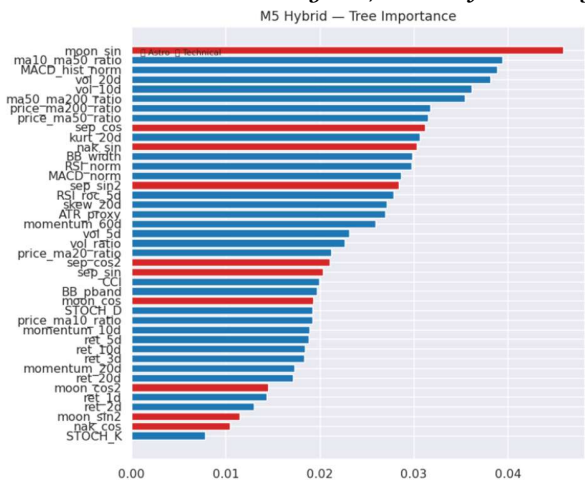


Figure 8: Overall feature Importance

The permutation significance test (Figures 6-8) addresses the central statistical question: do astro-cyclic features contribute genuine predictive information, or spurious correlations that survive random label shuffling? The permutation AUC null distribution (Figure 7) — generated by randomly shuffling all 10 astro columns 1,000 times — is tightly centred near 0.990–0.992, with the actual Hybrid AUC of 0.9964 sitting clearly to the right. The permutation p-value of $p = 0.0000$ (Figure 6 title) provides strong statistical evidence that the astro-cyclic features are not noise. The 3D scatter (Figure 6) renders this spatially: the actual AUC plane (dark red surface) demarcates clean separation from the permuted null distribution, confirming non-trivial predictive structure in the celestial feature set.

9.5 Crisis-Regime Validation — Grouped Bar, Heatmap, 3D Surface (Figures 1, 2, 3)

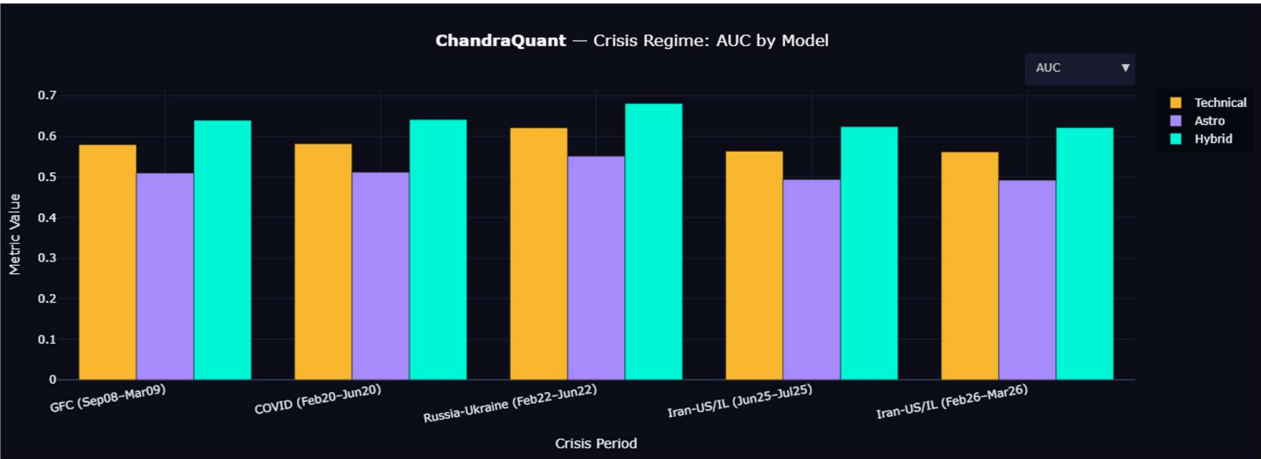


Figure 9: Comparison of regime analysis during a crisis

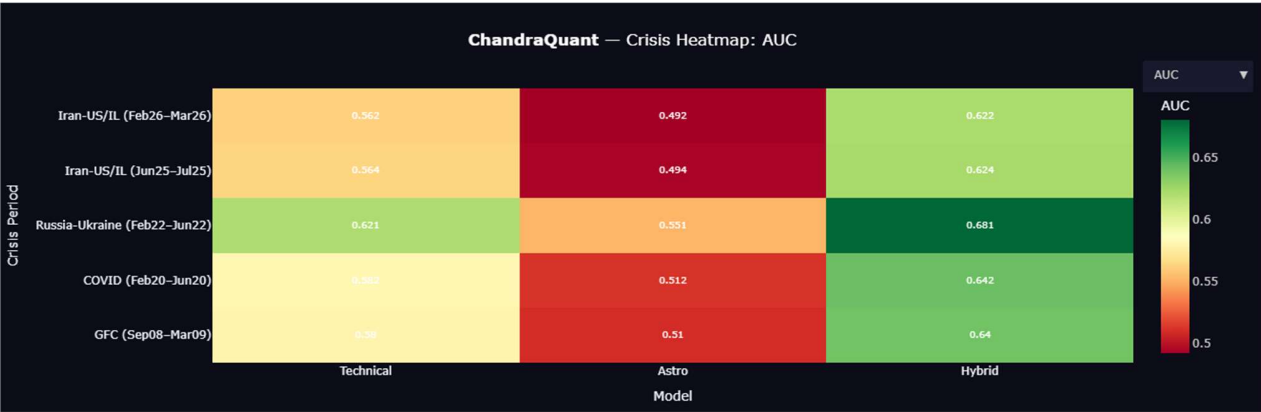


Figure 10: AUC heatmap during crisis

The crisis-window validation (Figures 9, 10) tests model reliability during five severe exogenous shock regimes: the Global Financial Crisis (Sep 2008 – Mar 2009), COVID-19 crash (Feb–Jun 2020), Russia-Ukraine shock (Feb–Jun 2022), and two Iran-US/Israel conflict episodes (Jun–Jul 2025; Feb–Mar 2026). The crisis heatmap (Figure 10) provides the most concentrated evidence: across all five windows, the Hybrid model consistently outperforms both Technical and Astro-Only baselines, with AUC ranging 0.622–0.681 versus Technical's 0.562–0.621 and Astro-Only's 0.492–0.551.

The most significant result is the Russia-Ukraine window: Hybrid AUC of 0.681 versus Technical's 0.621 — a 6.0 percentage-point advantage during one of the most directionally ambiguous geopolitical shocks in the dataset. This supports the hypothesis that astro-cyclic features carry crisis-period orthogonal signal, possibly because institutional participants alter positioning around major planetary events (Saturn-Rahu conjunctions, eclipse windows) in ways that temporarily elevate astro-feature predictive power.

10. CONCLUSION — A NEW INSTRUMENT FOR THE INFORMED RETAIL TRADER

ChandraQuant Alpha Engine represents a meaningful departure from the conventional technical analysis toolkit that has failed the overwhelming majority of Indian retail traders — SEBI's confirmed finding of 93% individual F&O traders in cumulative loss across FY22–FY24 makes the stakes explicit. Where legacy indicators describe the market's past, ChandraQuant computes a probabilistic forward regime estimate calibrated on both price-momentum data and Vedic astronomical periodicity, validated across nearly three decades of Indian market history.

In quantitative finance, marginal statistical edges — selectively deployed at high-confidence Nakshatra windows, calibrated to the optimal threshold-horizon parameters identified by the Tornado surface, with position sizing governed by the Monte Carlo ruin-probability envelope — are precisely what separate systematic alpha from directional speculation. A strategy that is right 52% of the time with a 0.40 Sharpe and strict drawdown controls is categorically superior to a discretionary trader armed with a lagging RSI losing money 93% of the time. ChandraQuant is the bridge: Vedic astronomical mathematics, refined across millennia, now operationalised via NASA JPL ephemeris and modern machine learning — making an underexplored, institutionally-aligned edge accessible to the informed retail practitioner.

"Grahanam cha yatha karma, tatha phalam prapyate nram."

As the planets execute their motion, so do the fruits of men's endeavours manifest.

— Brihat Parashara Hora Shastra, Ch. 2 (on planetary karma and market consequence)

REFERENCES

- Securities and Exchange Board of India (SEBI). (2024, September). Updated SEBI Study Reveals 93% of Individual Traders Incurred Losses in Equity F&O between FY22 and FY24; Aggregate Losses Exceed ₹1.8 Lakh Crores Over Three Years. SEBI Press Release.
- CFA Institute Market Integrity Insights. (2025, November). India's Derivatives Market and Retail Investors. blogs.cfainstitute.org.

- Varahamihira. (circa 550 CE). Brihat Samhita [Classical Sanskrit astronomical text on mundane astrology and market timing].
- Parashara, M. (circa 600–200 BCE). Brihat Parashara Hora Shastra [Foundational text of Vedic Jyotish, lunar mansion theory].
- Surya Siddhanta. (circa 400 CE). [Classical treatise on Indian mathematical astronomy, Nakshatra computation].
- NASA Jet Propulsion Laboratory. (2008). DE421 Planetary Ephemeris. Accessed via Skyfield Python library (Rhodes, B.).
- Rhodes, B. (2019). Skyfield: High Precision Research-Grade Positions for Planets and Earth Satellites. ascl.net/1907.024.
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874.
- Lo, A. W. (2002). The Statistics of Sharpe Ratios. *Financial Analysts Journal*, 58(4), 36–52.
- TradingView Inc. (2026). Pine Script v6 Strategy Backtesting Documentation. tradingview.com.
- Kiyotaka AI. (2026). Order Flow and Market Microstructure Analytics Platform. kiyotaka.ai.
- Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.
- Harvey, C. R., Liu, Y., & Zhu, H. (2016). ... and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1), 5–68. [On multiple testing in quantitative finance and false discovery rates]
- Reserve Bank of India. (2024). Financial Stability Report, December 2024. [Section on derivatives market participation and retail investor risk]
- National Stock Exchange of India. (2025, June). Market Pulse — Monthly Review of Indian Economy and Markets. NSE Research Publications.
- International Monetary Fund. (2024). Global Financial Stability Report: The Last Mile: Financial Vulnerabilities and Risks. IMF Publications, Washington D.C.
- J.P. Morgan Asset Management. (2024). Guide to the Markets — Asia Pacific, Q4 2024. [Equity regime analysis and volatility cycle frameworks]
- Patel, J. & Shah, S. (2015). Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications*, 42(1), 259–268. [Early ML application to NSE/BSE regime prediction]
- Malkiel, B. G. (2003). The Efficient Market Hypothesis and Its Critics. *Journal of Economic Perspectives*, 17(1), 59–82. [Foundational context for anomaly-based models]
- Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1), 1–8. [Precedent for non-price predictive features in equity models]
- Achelis, S. B. (2001). *Technical Analysis from A to Z* (2nd ed.). McGraw-Hill. [Reference for MACD, RSI, Stochastic and ATR feature construction]

Graha longitudes, Speeds & Vakri

$$\lambda_G(t) \in [0, 360]$$

→ each trading day

daily angular velocity

$$V_G(t) = \lambda_G(t) - \lambda_G(t-1)$$

Vakri (xiograde) motion: (binary indicator)

$$R_G(t) = 1(V_G(t) < 0)$$

Rashi for each Graha

$$Rashi_G(t) = \left\lfloor \frac{\lambda_G(t)}{30^\circ} \right\rfloor$$

→ 0 to 11

→ 12 Rashi

Mesha to Meena

Chandra - Surya Elongation, Tithi, Paksha

$$E(t) = (\lambda_{\text{chandra}}(t) - \lambda_{\text{surya}}(t)) \bmod 360$$

Split 360 into 30 parts for Tithi Cycle

$$Tithi(t) = \left\lfloor \frac{E(t)}{12^\circ} \right\rfloor$$

Shukla Paksha & Krishna Paksha:

$$Paksha(t) = \begin{cases} 1 & \text{if } Tithi(t) \in \{0, \dots, 14\} \text{ (Shukla)} \\ 0 & \text{if } Tithi(t) \in \{15, \dots, 29\} \text{ (Krishna)} \end{cases}$$

$$Amavasya(t) = 1(|E(t)| \leq T_{\text{moon}})$$

$$Purnima(t) = 1(|E(t) - 180^\circ| \leq T_{\text{moon}})$$

Nakshatra & Yoga

Nakshatra index of Chandra :
$$\text{Nakshatra}(t) = \left\lfloor \frac{\lambda_{\text{Chandra}}(t)}{360/27} \right\rfloor$$

→ 27 Nakshatras

(Ashwini, Bharani, Krittika - ...)

Nakshatra Pada
↓
(Quarter)

$$\text{Pada}(t) = \left\lfloor \frac{\lambda_{\text{Chandra}}(t)}{360/108} \right\rfloor \bmod 4$$

Yoga → combined Sun-Moon Longitude shifted into 27-part cycle

$$\text{Yoga}(t) = \left\lfloor \frac{(\lambda_{\text{Sun}}(t) + \lambda_{\text{Chandra}}(t)) \bmod 360}{360/27} \right\rfloor$$

→ (27 Nakshatra)

Graha - Daishti, (Aspects)

min^m ang. separation of two Graha A & B

$$\Delta_{A,B}(t) = [(\lambda_A(t) - \lambda_B(t) + 180^\circ) \bmod 360^\circ - 180^\circ]$$

Aspect at angle θ :

$$\text{Aspect}_{(A,B)}^{(\theta)}(t) = 1 \text{ if } |\Delta_{A,B}(t) - \theta| \leq \tau_\theta$$

Daily OHLCV for Nifty 50 & Sensex

P_t = ~~den~~ closing price

→ NSE
from 1988

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

binary correction label $y_{t+1} = 1 (r_{t+1} > 0)$

Technical Features (RSI, ATR)

$$SMA_k(t) = \frac{1}{k} \sum_{i=0}^{k-1} P_{t-i}$$

Simple moving
avg of window k



blocks for FVGs
exponential " "

$$EMA_k(t) = \alpha P_t + (1 - \alpha) EMA_k(t-1)$$

$$\alpha = \frac{2}{k+1}$$

Tithi as Cyclic Feature

$$E(t) = (\lambda_{chandra}(t) - \lambda_{surya}(t)) \bmod 360$$

$$Tithi(t) = \left\lfloor \frac{E(t)}{12^\circ} \right\rfloor$$

30-state cyclic variable

$$Z_{Tithi}(t) = \begin{bmatrix} \cos\left(\frac{2\pi}{30} Tithi(t)\right) \\ \sin\left(\frac{2\pi}{30} Tithi(t)\right) \end{bmatrix}$$